

Chapter 7 - Practice Set

Sets

Instructions: Write each answer as a separate Python file and run it to verify your output. Do not look at the Chapter 7 notes until you have attempted the problem.

Question 1

Write a program that creates a set of your five favourite movies (as strings). Then print the set, print its length using `len()`, check whether a specific movie is in the set using the `in` operator, and check whether another movie is not in the set using the `not in` operator. Finally, try to access the first element of the set using an index (e.g., `my_set[0]`) and observe the error that Python raises.

Question 2

Write a program that starts with an empty set (created using `set()`, NOT `{}`) and builds it up step by step: use `add()` to insert five different integers one at a time, then use `update()` to add three more integers from a list in a single call. Print the set after each modification to observe how it changes.

Question 3

Write a program that creates a set of at least six elements. Demonstrate all three removal methods on it: use `remove()` to delete an element that EXISTS in the set, use `discard()` to attempt to remove an element that does NOT exist in the set (and print a message confirming no error occurred), and use `pop()` to remove and print an arbitrary element. Print the set after each operation.

Question 4

Write a program that defines two sets: one representing students who passed Math, and one representing students who passed Science (use at least five student names in each, with some overlap). Use set operations to find and print: (a) all students who passed at least one subject (union), (b) students who passed both subjects (intersection), (c) students who passed Math but NOT Science (difference), and (d) students who passed exactly one subject but not both (symmetric difference). Use both the operator symbols and the method forms for at least two of these.

Question 5

Write a program that demonstrates the list-to-set-to-list conversion pattern. Start with a list of at least ten integers that contains several duplicate values. Convert it to a set to remove duplicates and print the set. Then convert the set back to a list, sort the list, and print the final sorted list with no duplicates. Also print the length of the original list and the length of the final list to show how many duplicates were removed.

Question 6

Write a program that creates a frozenset from a regular set of your choice. Show that you can still use `len()`, the `in` operator, and a set operation (such as intersection with another frozenset or set) on it. Then deliberately attempt to call `add()` on the frozenset and observe the `AttributeError` that Python raises. Finally, use the frozenset as a key in a dictionary and store a value for it, then print the dictionary.

Question 7

Write a program that uses set comprehension to create the following sets: (a) the set of all even numbers from 1 to 20, (b) the set of all squared values of numbers from -5 to 5 (notice how the result has fewer elements than the input range because squaring negatives and positives gives the same value), and (c) the set of the lengths of each word in the sentence "the quick brown fox jumps over the lazy dog" (you will get the set of unique word lengths). Print each resulting set.

Question 8

Write a program that reads a sentence from the user using `input()` and finds all the unique characters in it (ignoring spaces). Store these characters in a set. Then create a second set containing all the vowels (a, e, i, o, u). Use set operations to print: (a) which vowels appear in the user's sentence, and (b) which characters in the user's sentence are NOT vowels. Use the `in` operator to check if the letter "e" appears in the sentence.

Question 9

Write a program that creates two sets representing the tags or genres of two different playlists (for example, {"pop", "rock", "indie"} and {"jazz", "rock", "blues"}). Use set methods and operations to: (a) add two new tags to the first playlist using `update()`, (b) remove a tag from the second playlist using `discard()`, (c) merge both playlists into one combined set using `union`, (d) find tags that are exclusive to each playlist using `symmetric difference`. Print the state of both playlists and all results.

Question 10

Write a program that demonstrates three of the common beginner mistakes covered in Chapter 7, each handled gracefully using a try-except block. Mistake (a): Create a variable using {} and show it is a dict, not a set, then create the correct empty set using set(). Mistake (b): Try to access a set element by index inside a try block, catch the TypeError, and print a message explaining why it fails. Mistake (c): Try to add a list (e.g., [1, 2, 3]) into a set inside a try block, catch the TypeError, then fix it by adding a tuple instead. Print clear messages at each step explaining what is happening.

--- End of Chapter 7 Practice Set ---